ARCHITECTURE OF A DATABASE MANAGEMENT SYSTEM (DBMS)

Technical Presentation | CO1 Assessment Tool 3

SIMATS ENGINEERING

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Course Code: CSA05 | Course: Database Management Systems

1. What Is a DBMS?

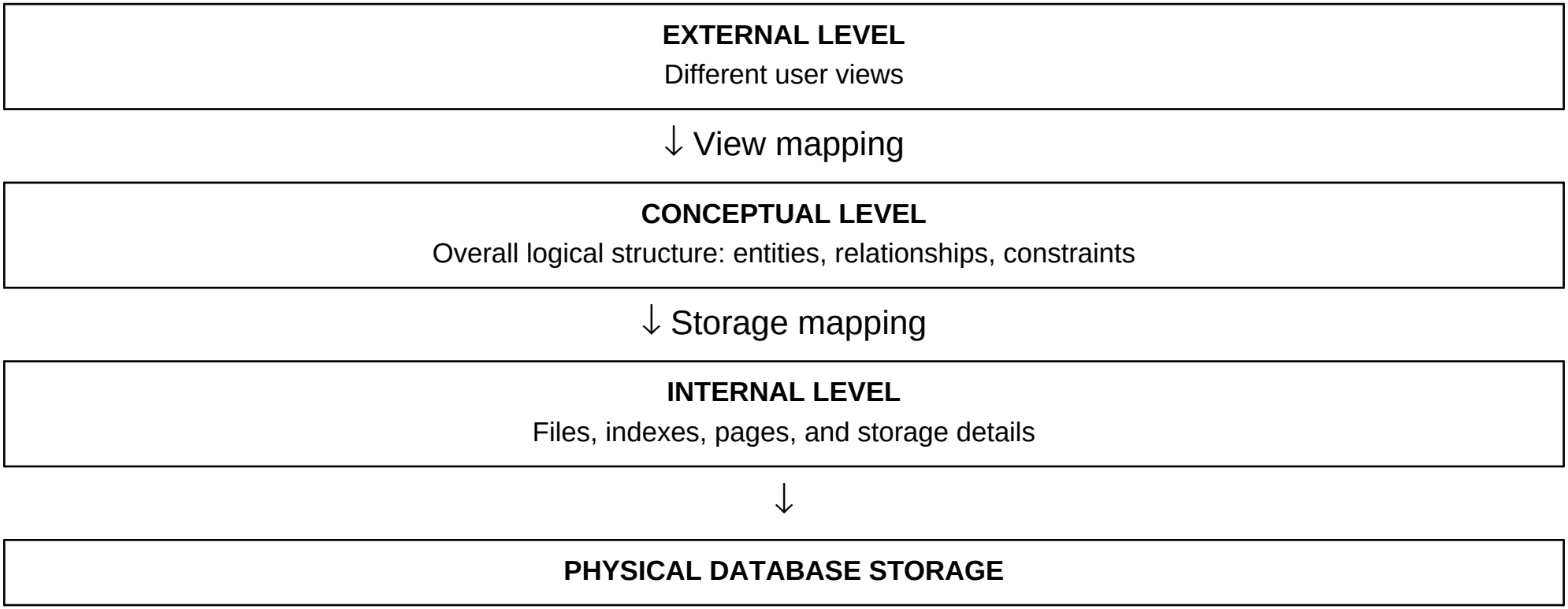
- **Database:** an organized collection of related data.
- **DBMS:** software used to define, store, retrieve, update, and control data.
- It acts as an interface between users/applications and the stored database.
- **Examples:** MySQL, PostgreSQL, Oracle Database, and Microsoft SQL Server.
- **Why it matters:** helps manage large amounts of data consistently and securely.

2. The Problem a DBMS Solves

- **File-based approach:** separate files can create duplicate data, inconsistent updates, and difficult sharing.
- **DBMS approach:** central management supports controlled access, querying, backup, and recovery.
- **Example:** a college needs consistent student, course, attendance, and marks records.
- **Engineering goal:** improve reliability, integrity, availability, and controlled data access.

3. Three-Level DBMS Architecture

The three-schema architecture separates user views from the logical database design and physical storage.



Benefit: changes to physical storage can often be made without changing every user view.

4. Main Functional Components

- **Query processor:** parses and optimizes SQL queries.
- **Storage manager:** organizes data files, indexes, and buffers.
- **Transaction manager:** coordinates transactions and concurrency.
- **Recovery manager:** helps restore consistency after failures.
- **Authorization and integrity:** enforce permissions and data constraints.

5. How a Query Is Processed

- 1. User/application submits an SQL statement.
- 2. Parser checks syntax and interprets the query.
- 3. Optimizer chooses an efficient execution plan.
- 4. Execution engine requests records through the storage manager.
- 5. Results are returned to the user/application.

SQL QUERY	→ **PARSE**	→ **OPTIMIZE**	→ **EXECUTE**	→ **RESULTS**

6. Transactions, Security, and Recovery

- **Transactions:** group operations into one logical unit; ACID properties support reliable processing.
- **Concurrency control:** coordinates simultaneous users to prevent conflicting updates.
- **Security:** roles and privileges limit who can view or change data.
- **Backup and recovery:** logs and backups help recover from errors or failures.
- **Example:** student marks should not be partly updated or visible to unauthorized users.

7. Case Study: College Database

- **Entities:** Student, Course, Faculty, Enrollment, Assessment.
- **Relationships:** a student enrolls in courses; faculty teach courses; assessments record results.
- **Possible controls:** unique student ID, valid course ID, marks within an allowed range, role-based access.
- **DBMS benefit:** less duplication, reliable reports, controlled access, and easier backup.

8. DBMS Architecture: Strengths and Trade-offs

- **Strengths:** data independence, shared access, security, integrity, and recovery.
- **Trade-offs:** setup and administration costs, resource overhead, and need for skilled management.
- **File system:** may suit small, simple, isolated data needs.
- **DBMS:** is better when data is shared, queried frequently, protected, and kept consistent.
- **Design principle:** choose architecture based on scale, risk, access patterns, and requirements.

9. Summary & References

Summary: DBMS architecture separates user views, logical design, and physical storage.

Its components process queries, manage storage and transactions, enforce security, and support recovery.

The architecture helps organizations maintain reliable, shared, and controlled data.

References

1. Elmasri, R. and Navathe, S. B., *Fundamentals of Database Systems*, Pearson.

2. Silberschatz, A., Korth, H. F., and Sudarshan, S., *Database System Concepts*, McGraw-Hill.

3. PostgreSQL Documentation, “Architecture” and “SQL Commands,” https://www.postgresql.org/docs/

Thank you — Questions?

DATABASE MANAGEMENT SYSTEMS | CSA05

Evaluation Rubrics: How This Presentation Targets Level 4

Criterion	Weight	What to demonstrate
Technical Content Accuracy	30%	Define DBMS and explain architecture, components, transactions, and recovery accurately.
Problem Identification	25%	Explain file-system problems and analyze a college database case with a logical engineering approach.
Use of Tools	20%	Use the three-level architecture diagram and query-processing flow; explain each visual clearly.
Communication	15%	Speak clearly, follow the slide order, use examples, and finish in about 8 minutes.
Professionalism	10%	Use original explanations, cite references, maintain formal delivery, and answer questions honestly.

Suggested timing: Slides 1–2 (1.5 min) • Slide 3 (1 min) • Slides 4–6 (2.5 min) • Slides 7–8 (1.5 min) • Slide 9 (1 min) • buffer (0.5 min).

Before submitting: fill in your name and register number; review the college's academic-integrity requirements and explain the content in your own words.